Note on Visakhapatnam Port Authority's Achievement in the CPPI 2023 Rankings

In a historic development for India's maritime sector, Visakhapatnam Port Authority (VPA) has been ranked 19th globally in the Container Port Performance Index (CPPI) 2023, a joint publication by the World Bank and S&P Global Market Intelligence. This marks a dramatic improvement from its 115th rank in 2022 and makes it the first Indian port ever to enter the Global Top 20. This significant leap is a direct result of VPA's focused efforts on infrastructure augmentation, operational streamlining, digital integration, and sustainability initiatives.

The CPPI evaluates over 400 container ports worldwide based on critical metrics such as operational efficiency, vessel turnaround time, and crane productivity. This achievement is part of a broader national milestone, as nine Indian ports were ranked in the Global Top 100 for the first time, an outcome hailed by Union Minister for Ports, Shipping & Waterways, Shri Sarbananda Sonowal, as a direct result of the Sagarmala Programme under the leadership of Prime Minister Narendra Modi.

Port Name	Overall Ranking	Port Name	Overall Ranking
YANGSHAN	1	VISAKHAPATNAM	19
SALALAH	2	YEOSU	20
CARTAGENA (COLOMBIA)	3	TIANJIN	21
TANGER-MEDITERRANEAN	4	YANTIAN	22
TANJUNG PELEPAS	5	TANJUNG PRIOK	23
CHIWAN	6	LIANYUNGANG	24
CAI MEP	7	SHEKOU	25
GUANGZHOU	8	CALLAO	26
YOKOHAMA	9	MUNDRA	27
ALGECIRAS	10	PORT KLANG	28
HAMAD PORT	11	KHALIFA PORT	29
NINGBO	12	KING ABDULLAH PORT	30
MAWAN	13	XIAMEN	31
DALIAN	14	BUSAN	32
HONG KONG	15	GEMLIK	33
PORT SAID	16	BARCELONA	34
SINGAPORE	17	DAMMAM	35
KAOHSIUNG	18	SAVONA-VADO	36

Figure 1: The CPPI 2023: Global Ranking of the Container Ports

In the CPPI 2023 global rankings, Visakhapatnam ranked 19th, just behind Kaohsiung (18th) and Singapore (17th), while outperforming major global players like Mundra (27th) and Port Klang (28th) *(Fig.1).* This places Visakhapatnam ahead of several long-established global transshipment hubs, reflecting its emergence as a globally competitive container port.

Unlike perception-based indices, the CPPI strictly evaluates empirical time-based metrics such as arrival-to-berth duration, berth idle time, and gross crane productivity, offering a highly credible and performance-driven assessment. The CPPI 2023 assessed 194,198 port calls, covering 253.7 million container moves across 508 ports and 876 terminals globally. This reinforces the credibility of Visakhapatnam's achievement amid a large and competitive peer group.

The operational success of Visakhapatnam Port stems largely from the performance of Visakha Container Terminal Pvt. Ltd. (VCTPL), operated by JM Baxi Ports & Logistics. The terminal achieved a Gross Crane Rate (GCR) of 27.9 moves per crane hour, an average turnaround time of 19.1 hours, and a pre-berthing delay of only 1.9 hours in FY 2023–24. In contrast, the global average port call duration in 2023 was 40.5 hours, with 11.7% (approximately 3.7 hours) as unproductive berth idle time. Visakhapatnam's figures highlight its high performance relative to global standards.

This represents a marked improvement over previous year. In FY 2021–22, VCTPL recorded a GCR of 23.3 moves per hour, an average turnaround time of 30.95 hours, and a pre-berthing delay of 0.91 hours. During the same period, the average port call duration was 27.7 hours. The progress from FY 2021–22 to FY 2023–24 showcases a clear trajectory of operational enhancement through targeted interventions and capacity upgrades.

A major enabler of these outcomes has been VPA’s strategic infrastructure development, which has evolved substantially in recent years and is a key driver behind the port’s improved CPPI ranking. Prior to 2022, the port operated with a modest setup of 4 Post Panamax single-lift quay cranes and 10 Rubber-Tyred Gantry Cranes (RTGCs). However, recognizing the growing demand and the need for faster, more efficient operations, VPA undertook major infrastructure upgrades.

In December 2021, the quay at Terminal-1 was extended from 450 meters to 895 meters, making it the longest linear quay on India’s East Coast. With a draft of 16.5 meters, it allows berthing of container vessels up to 390 meters LOA (Length Overall). Simultaneously, the port deployed three Super Post Panamax RMQCs (Rail Mounted Quay Cranes), each equipped with 65-meter outreach and 65-ton SWL (Safe Working Load) in addition to the existing four cranes, increasing the total to seven quay cranes. The number of RTGCs also rose from 10 to 19, enabling faster container movement and better yard-side handling. These infrastructure advancements, particularly the adoption of twin-lift crane technology and extended berths directly contributed to reduced vessel turnaround times, higher crane productivity, and ultimately VPA’s rise in the CPPI rankings.

Figure 2: Three Super Post Panamax quay cranes deployed for efficient ship-to-shore container handling operations.

The container storage yard has also been upgraded with an additional 41 acres, enabling 2,600 more ground slots and raising the daily handling capacity to 18,000 TEUs (Twenty Foot Equivalent Units), thereby reducing congestion and improving yard productivity. To facilitate seamless cargo evacuation, a new 6-lane kiosk-enabled gate complex has been commissioned in August 2022, alongside the existing 3-lane facility, streamlining trailer movement and reducing turnaround time for cargo vehicles.

In line with India's commitment to sustainability and the Maritime India Vision 2030, VPA has adopted a green and future-ready operational framework. It currently operates a 10 MW solar power plant since 2016, with plans to expand by another 15 MW to move toward complete solar dependency. Over 5.65 lakh trees have been planted on 650 acres as part of an afforestation initiative, contributing to ecological restoration. Terminal-2 is equipped with 9 electric Rubber-Tyred Gantry Cranes (RTGCs)*(Fig.3)*, fitted with GPS-based position detection, in addition to 10 electric RTGCs at Terminal-1, helping reduce the port's carbon footprint.

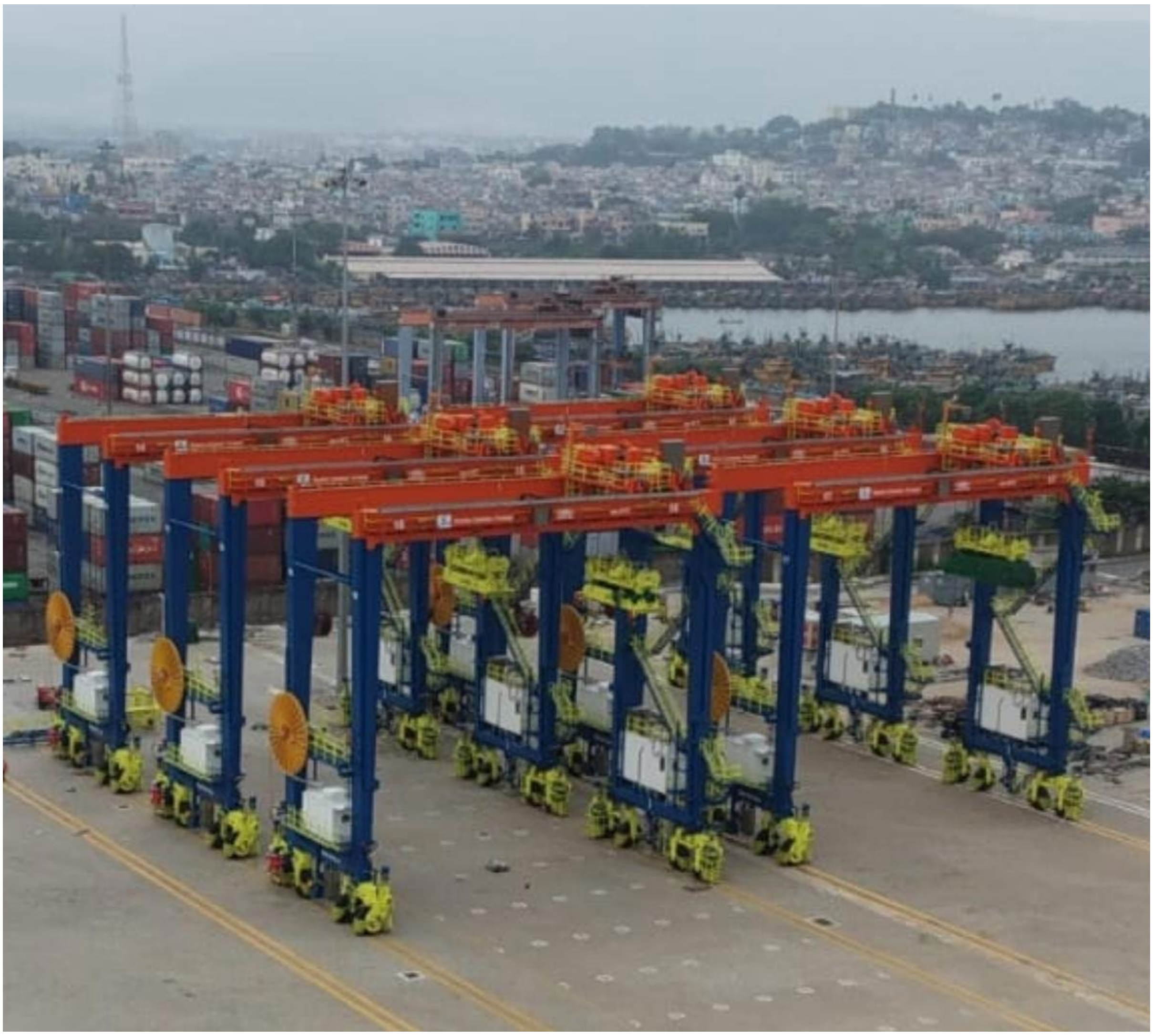
Figure 3: Nine Rubber-Tyred Gantry Cranes (RTGCs) deployed for efficient yard-side container handling operations.

Visakhapatnam's strategic location near the Malacca Strait and strong rail-road connectivity position it as a key gateway to the hinterlands of Andhra Pradesh, Telangana, Odisha, Chhattisgarh, Maharashtra, and more. It also serves as the second gateway for Nepal's EXIM trade. The port has also integrated advanced safety and automation systems. Both terminals are fully covered by CCTV and Wi-Fi networks, allowing real-time monitoring and minimal human-machine interface. Other sustainability features include sewage treatment plants, firefighting systems, and clean power infrastructure. For refrigerated cargo, Terminal-2 has added 300 new reefer plug points, backed by round-the-clock DG support, supplementing the 350 plug points already in place at Terminal-1. These enhancements support India's growing marine export trade.

Collaboration with stakeholders including Customs, CONCOR, Indian Railways, and over 65 container lines has ensured ecosystem-level efficiency. These infrastructure and operational reforms have led to record cargo performance. In FY 2023–24 (as of March 27), VPA handled 80.5 million metric tonnes (MMT) of cargo, surpassing the previous record of 73.75 MMT in

FY 2022–23. Key growth segments included petroleum products, iron ore, fertilizers, and containerized cargo. VPA's growth aligns with national priorities under the PM Gati Shakti National Master Plan, the National Logistics Policy, and Sagarmala, and contributes to India's aspiration of becoming a top global logistics hub.

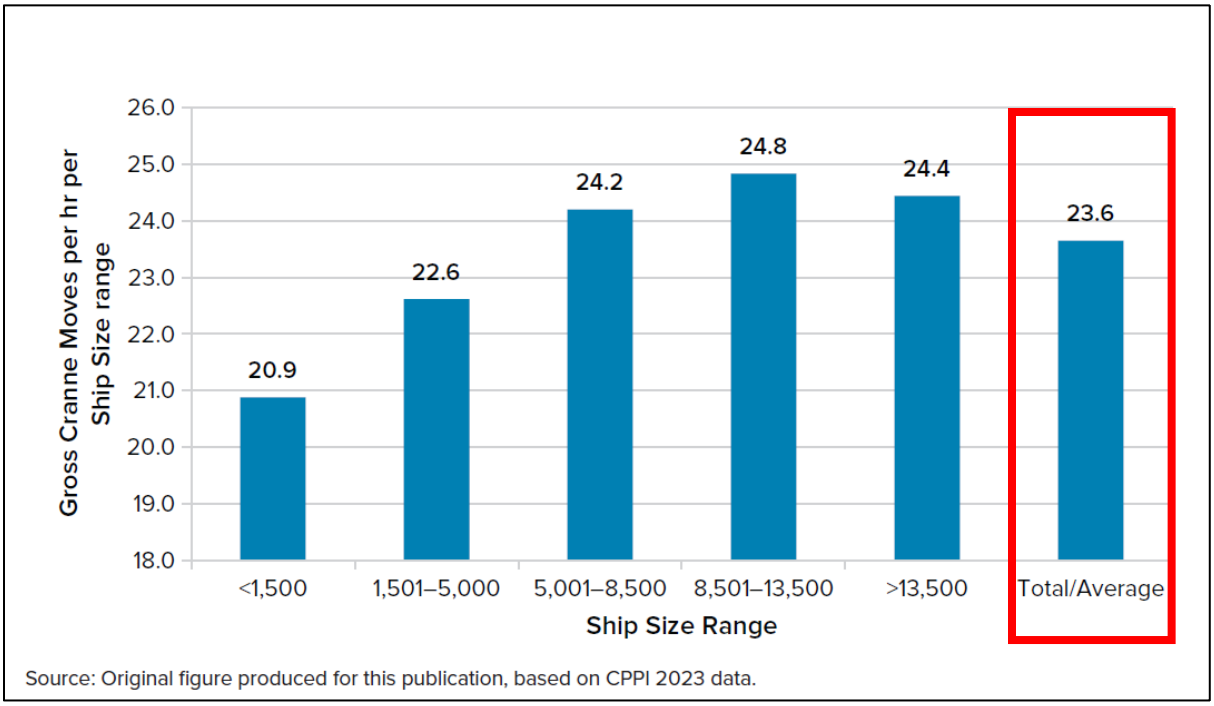

Figure 4: Container Moves Performed per gross Crane Hour across Various Ship Sizes

Visakhapatnam's performance is underpinned by real-time data systems like AIS tracking, used in CPPI methodology. This strengthens the port's case as a replicable model. Moreover, the global average crane productivity of 23.5 moves/hour was significantly exceeded by Visakhapatnam's 27.9 moves/hour—a ~18% higher productivity rate (*Fig.2*).

In conclusion, the transformation of Visakhapatnam Port Authority from a regional player to a globally competitive container port is a compelling case of how strategic investments, modern infrastructure, efficient governance, and sustainability initiatives can elevate port performance. As VPA targets over 1 million TEUs (Twenty Foot Equivalent Units) annually by 2026, it is poised to emerge as a container hub on the East Coast of India. Its CPPI success exemplifies the impact of targeted infrastructure investment, stakeholder coordination, digitalization, and sustainability. The VPA model offers valuable lessons for other Indian ports and logistics gateways and presents a scalable and replicable framework for port-led development. This success story is a suitable candidate for national capacity building initiatives under NITI Aayog and supports the broader goal of improving India's ranking in global logistics performance indices.